Group 2

The code

```
89      -- سيلكت من جدول بارتي
90      SELECT PartyID, PartyType, Location
        FROM Partyy
        WHERE PartyType = 'Birthday';
93
94      -- سيلكت من جدول كستمَر
        SELECT CustomerID, Name
        FROM Customerr
97      WHERE Name LIKE 'A%';
98
        -- سيلكت من جدول ستاف
        SELECT StaffID, Name, Role, PartyID
101     FROM Staff
102     WHERE Role = 'Manager';
103
        -- سيلكت من جدول تيكيت
        SELECT TicketID, CustomerID, Price, Part
106     FROM Ticket
107     WHERE Price > 20;
108
        -- سيلكت من جدول ارقام العملاء
        SELECT CustomerID, Phone
111     FROM Customer_Phone
112     WHERE CHAR_LENGTH(Phone) = 10;
113
        SELECT COUNT(*) AS TotalCustomers
        FROM Customerr;
116
```

```
SELECT SUM(Price) AS TotalRevenue
FROM Ticket;


SELECT AVG(Price) AS AVG_Price
FROM Ticket;


SELECT PartyType, Location, COUNT(*) AS TotalParties
FROM Partyy
GROUP BY PartyType, Location;



CREATE TABLE Customer (
    CustomerID INT,
    Name VARCHAR(100)
);

CREATE TABLE Party (
    PartyID INT,
    PartyType VARCHAR(50),
    Location VARCHAR(100)
);

CREATE TABLE Ticket (
    TicketID INT,
    PartyID INT,
```

The code

```
    CustomerID INT
);

INSERT INTO Customer VALUES
(1, 'Sara'),
(2, 'Lama'),
(3, 'Reema');

INSERT INTO Party VALUES
(1, 'Birthday', 'Jeddah'),
(2, 'Wedding', 'Riyadh');

INSERT INTO Ticket VALUES
(1, 1, 1),
(2, 1, 1),
(3, 2, 2);

-- =========  JOIN =========
SELECT
    Customer.Name AS CustomerName,
    Party.PartyType AS EventType,
    Party.Location
FROM Ticket
INNER JOIN Customer
    ON Ticket.CustomerID = Customer.CustomerID
INNER JOIN Party
    ON Ticket.PartyID = Party.PartyID;
```

```
-- =========  Subquery =========
SELECT Name
FROM Customer
WHERE CustomerID IN (
    SELECT CustomerID
    FROM Ticket
    GROUP BY CustomerID
    HAVING COUNT(TicketID) > 1
);

CREATE VIEW Simple_Customers AS
SELECT *
FROM Customerr;
SELECT * FROM Simple_Customers;
```

The Code & The Output

```
83  SELECT * FROM Customerr;
84  SELECT * FROM Staff;
85  SELECT * FROM Ticket;
86  SELECT * FROM Customer_Phone;
87
88  -- سلكت من جدول بارتي
89  SELECT PartyID, PartyType, Location
90  FROM Partyy
91  WHERE PartyType = 'Birthday';
92
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

PartyID	PartyType	Location
1	Birthday	Riyadh - Hall A
5	Birthday	Khobar - Private Hall
NULL	NULL	NULL

Result Grid | Form Editor | Apply | Context Help | Snippe

Partyy 5

Output

Action Output

#	Time	Action	Message	Dura
12	00:09:42	SELECT CustomerID, Name FROM Customerr WHERE Name LIKE 'A%' LIMIT...	1 row(s) returned	0.00
13	00:09:50	SELECT TicketID, CustomerID, Price, PartyID FROM Ticket WHERE Price > ...	2 row(s) returned	0.00
14	00:12:15	USE PartyManagement	0 row(s) affected	0.00
15	00:12:23	SELECT PartyID, PartyType, Location FROM Partyy WHERE PartyType = 'Birt...	2 row(s) returned	0.01

Select from Party

The Code & The Output

```
SELECT PartyID, PartyType, Location
FROM Partyy
WHERE PartyType = 'Birthday';

-- سيلكت من جدول كستمر
SELECT CustomerID, Name
FROM Customerr
WHERE Name LIKE 'A%';

-- سيلكت من جدول ستاف
```

CustomerID	Name
2	Ali Khaled
NULL	NULL

#	Time	Action	Message
13	00:09:50	SELECT TicketID, CustomerID, Price, PartyID FROM Ticket WHERE Price > ...	2 row(s) returned
14	00:12:15	USE PartyManagement	0 row(s) affected
15	00:12:23	SELECT PartyID, PartyType, Location FROM Partyy WHERE PartyType = 'Birt...	2 row(s) returned
16	00:13:01	SELECT CustomerID, Name FROM Customerr WHERE Name LIKE 'A%' LIMIT...	1 row(s) returned

Select from Customer table

The Code & The Output

```
95    FROM Customerr
96    WHERE Name LIKE 'A%';
97
98    -- سلکت من جدول ستاف
99    SELECT StaffID, Name, Role, PartyID
100   FROM Staff
101   WHERE Role = 'Manager';
102
103   -- سلکت من جدول تیکت
104   SELECT TicketID, CustomerID, Price, PartyID
```

Result Grid

StaffID	Name	Role	PartyID
1	Mais	Manager	1
2	Ali	Manager	2
NULL	NULL	NULL	NULL

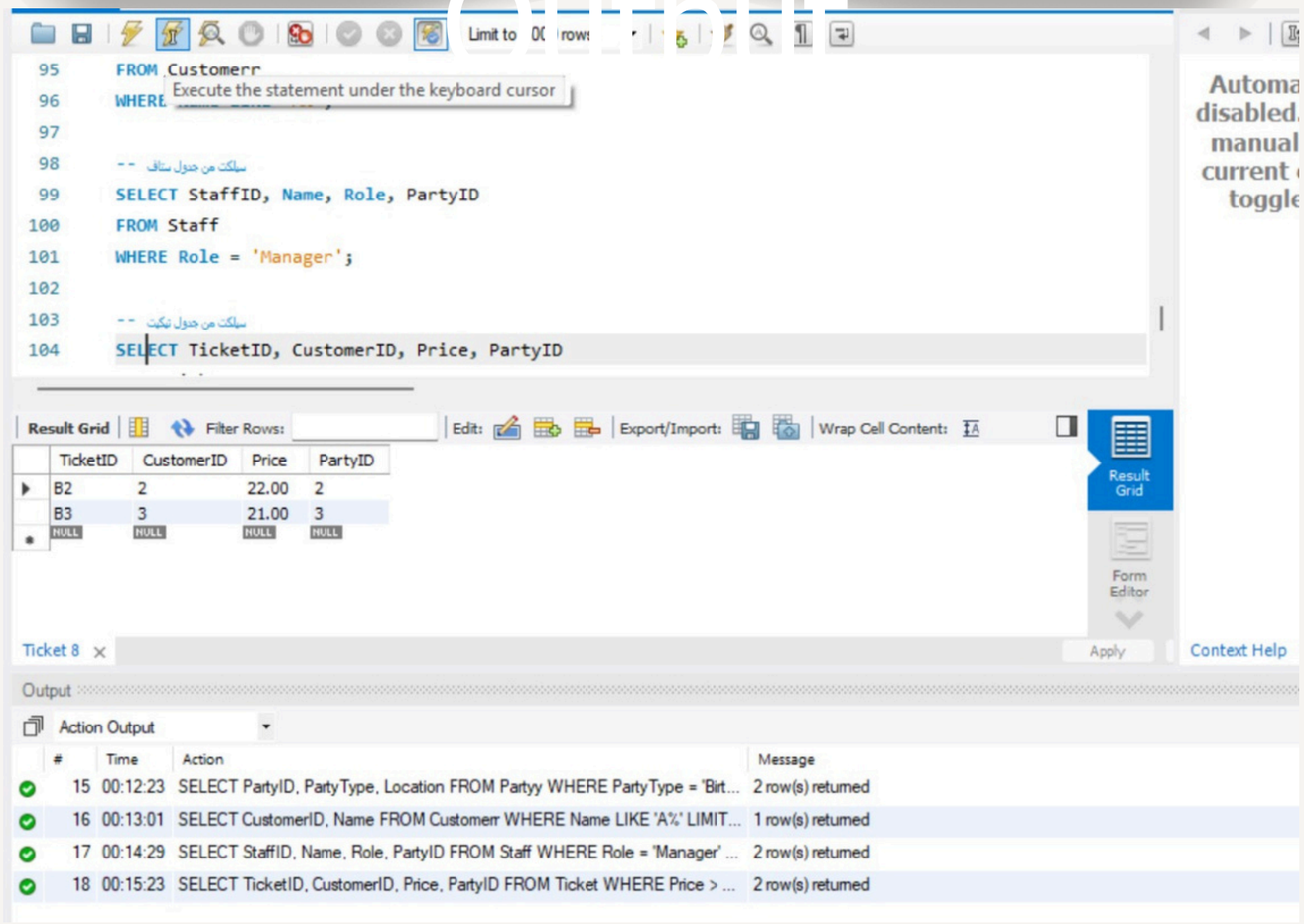
Output

Action Output

#	Time	Action	Message	Dur
14	00:12:15	USE PartyManagement	0 row(s) affected	0.00
15	00:12:23	SELECT PartyID, PartyType, Location FROM Partyy WHERE PartyType = 'Birt...	2 row(s) returned	0.01
16	00:13:01	SELECT CustomerID, Name FROM Customerr WHERE Name LIKE 'A%' LIMIT...	1 row(s) returned	0.00
17	00:14:29	SELECT StaffID, Name, Role, PartyID FROM Staff WHERE Role = 'Manager' ...	2 row(s) returned	0.01

Select from Staff table

The Code & The Output



Select from Ticket table

The Code & The Output

```
103    -- سيلكت من جدول تيكيت
104    SELECT TicketID, CustomerID, Price, PartyID
105    FROM Ticket
106    WHERE Price > 20;
107
108    -- سيلكت من جدول ارقام العملاء
109    SELECT CustomerID, Phone
110    FROM Customer_Phone
111    WHERE CHAR_LENGTH(Phone) = 9;
112
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

CustomerID	Phone
1	054367943
2	058765432
3	053217645
4	052487654
5	054582755
NULL	NULL

Result Grid | Form Editor | Apply

Customer_Phone 9

Output

Action Output

#	Time	Action	Message
16	00:13:01	SELECT CustomerID, Name FROM Customer WHERE Name LIKE 'A%' LIMIT...	1 row(s) returned
17	00:14:29	SELECT StaffID, Name, Role, PartyID FROM Staff WHERE Role = 'Manager' ...	2 row(s) returned
18	00:15:23	SELECT TicketID, CustomerID, Price, PartyID FROM Ticket WHERE Price > ...	2 row(s) returned
19	00:15:52	SELECT CustomerID, Phone FROM Customer_Phone WHERE CHAR_LENG...	5 row(s) returned

Select from

The Code & The Output

```
111  FROM Customer_Phone
112  WHERE CHAR_LENGTH(Phone)
113
114  SELECT COUNT(*) AS TotalCustomers
115  FROM Customerr;
116
117  SELECT SUM(Price) AS TotalRevenue
118  FROM Ticket;
119
120
121  SELECT AVG(Price) AS AVG_Price
122  FROM Ticket;
123
124
125  SELECT PartyType, Location, COUNT(*) AS TotalParties
126  FROM Partyy
127  GROUP BY PartyType, Location;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

TotalCustomers
5

Result 21

Output

Action Output

#	Time	Action	Message
36	11:14:25	SELECT * FROM Ticket LIMIT 0, 1000	5 row(s) returned
37	12:20:58	SELECT COUNT(*) AS TotalCustomers FROM Customerr LIMIT 0, 1000	1 row(s) returned
38	12:21:07	SELECT SUM(Price) AS TotalRevenue FROM Ticket LIMIT 0, 1000	1 row(s) returned
39	12:21:15	SELECT AVG(Price) AS AVG_Price FROM Ticket LIMIT 0, 1000	1 row(s) returned
40	12:21:21	SELECT PartyType, Location, COUNT(*) AS TotalParties FROM Partyy GROUP BY PartyType, Location LIMI...	5 row(s) returned
41	12:23:42	SELECT COUNT(*) AS TotalCustomers FROM Customerr LIMIT 0, 1000	1 row(s) returned

Count total customers

The Code & The Output

```
104    -- سيلكت من جدول تيكت
105 •  SELECT TicketID, Customer ID, Price, Part ID
106    FROM Ticket
107    WHERE Price > 20;
108
109    -- سيلكت من جدول ارقام العملاء
110 •  SELECT CustomerID, Phone
111    FROM Customer_Phone
112    WHERE CHAR_LENGTH(Phone) = 9;
113
114 •  SELECT COUNT(*) AS TotalCustomers
115    FROM Customerr;
116
117 •  SELECT SUM(Price) AS TotalRevenue
118    FROM Ticket;
119
120
121 •  SELECT AVG(Price) AS AVG_Price
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

TotalRevenue
94.00

Result 22

Output

Action Output

#	Time	Action	Message
37	12:20:58	SELECT COUNT(*) AS TotalCustomers FROM Customerr LIMIT 0, 1000	1 row(s) returned
38	12:21:07	SELECT SUM(Price) AS TotalRevenue FROM Ticket LIMIT 0, 1000	1 row(s) returned
39	12:21:15	SELECT AVG(Price) AS AVG_Price FROM Ticket LIMIT 0, 1000	1 row(s) returned
40	12:21:21	SELECT PartyType, Location, COUNT(*) AS TotalParties FROM Partyy GROUP BY PartyType, Location LIMI...	5 row(s) returned
41	12:23:42	SELECT COUNT(*) AS TotalCustomers FROM Customerr LIMIT 0, 1000	1 row(s) returned

Sum total revenue

The Code & The Output

```
113
114 •  SELECT COUNT(*) AS Tota Cust ers
115    FROM Customerr;
116
117 •  SELECT SUM(Price) AS TotalRevenue
118    FROM Ticket;
119
120
121 •  SELECT AVG(Price) AS AVG_Price
122    FROM Ticket;
123
124
125 •  SELECT PartyType, Location, COUNT(*) AS TotalParties
126    FROM Partyy
127    GROUP BY PartyType, Location;
128
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

AVG_Price
18.800000

Result 23

Output

Action Output

#	Time	Action	Message
38	12:21:07	SELECT SUM(Price) AS TotalRevenue FROM Ticket LIMIT 0, 1000	1 row(s) returned
39	12:21:15	SELECT AVG(Price) AS AVG_Price FROM Ticket LIMIT 0, 1000	1 row(s) returned
40	12:21:21	SELECT PartyType, Location, COUNT(*) AS TotalParties FROM Partyy GROUP BY PartyType, Location LIMI...	5 row(s) returned
41	12:23:42	SELECT COUNT(*) AS TotalCustomers FROM Customerr LIMIT 0, 1000	1 row(s) returned

Average ticket price

The Code & The Output

```sql
SELECT COUNT(*) AS Tota Custome s
FROM Customerr;

SELECT SUM(Price) AS TotalRevenue
FROM Ticket;


SELECT AVG(Price) AS AVG_Price
FROM Ticket;


SELECT PartyType, Location, COUNT(*) AS TotalParties
FROM Partyy
GROUP BY PartyType, Location;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

PartyType	Location	TotalParties
Birthday	Riyadh - Hall A	1
Wedding	Jeddah - Beach Resort	1
Corporate	Dammam - Business Center	1
Concert	Riyadh - Boulevard Arena	1
Birthday	Khobar - Private Hall	1

Result 24

Output

Action Output

#	Time	Action	Message
39	12:21:15	SELECT AVG(Price) AS AVG_Price FROM Ticket LIMIT 0, 1000	1 row(s) returned
40	12:21:21	SELECT PartyType, Location, COUNT(*) AS TotalParties FROM Partyy GROUP BY PartyType, Location LIMI...	5 row(s) returned
41	12:23:42	SELECT COUNT(*) AS TotalCustomers FROM Customerr LIMIT 0, 1000	1 row(s) returned
42	12:24:40	SELECT SUM(Price) AS TotalRevenue FROM Ticket LIMIT 0, 1000	1 row(s) returned
43	12:25:06	SELECT AVG(Price) AS AVG_Price FROM Ticket LIMIT 0, 1000	1 row(s) returned

Number fo parties by type

The Code

```sql
-- ========= إنشاء الجداول =========

CREATE TABLE Customer (
    CustomerID INT,
    Name VARCHAR(100)
);

CREATE TABLE Party (
    PartyID INT,
    PartyType VARCHAR(50),
    Location VARCHAR(100)
);

CREATE TABLE Ticket (
    TicketID INT,
    PartyID INT,
    CustomerID INT
);

-- ========= إدخال بيانات بسيطة =========

INSERT INTO Customer VALUES
(1, 'Sara'),
(2, 'Lama'),
(3, 'Reema');

INSERT INTO Party VALUES
(1, 'Birthday', 'Jeddah'),
(2, 'Wedding', 'Riyadh');

INSERT INTO Ticket VALUES
(1, 1, 1),
(2, 1, 1),
```

This section shows the created tables and sample data used to support the JOIN and SUPQUERY operations performed later

The Code

```
(2, 1, 1),
(3, 2, 2);

-- =========  JOIN =========
SELECT
    Customer.Name AS CustomerName,
    Party.PartyType AS EventType,
    Party.Location
FROM Ticket
INNER JOIN Customer
    ON Ticket.CustomerID = Customer.CustomerID
INNER JOIN Party
    ON Ticket.PartyID = Party.PartyID;

-- =========  Subquery =========
SELECT Name
FROM Customer
WHERE CustomerID IN (
    SELECT CustomerID
    FROM Ticket
    GROUP BY CustomerID
    HAVING COUNT(TicketID) > 1
);
```

This step performs a JOIN to link the tables and a SUPQUERY to identify customers with more than one ticket

The Output

STDIN

Input for the program (Optional)

Output:

```
+--------------+-----------+----------+
| CustomerName | EventType | Location |
+--------------+-----------+----------+
| Sara         | Birthday  | Jeddah   |
| Sara         | Birthday  | Jeddah   |
| Lama         | Wedding   | Riyadh   |
+--------------+-----------+----------+
+------+
| Name |
+------+
| Sara |
+------+
```

Join and Supquery opreation

The Code & The Output

```
179     SELECT CustomerID
180     FROM Ticket
181     GROUP BY CustomerID
182     HAVING COUNT(TicketID) > 1
183 );
184
185 • CREATE VIEW Simple_Customers AS
186   SELECT *
187   FROM Customerr;
188 • SELECT * FROM Simple_Customers;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

CustomerID	Name
1	Sara Ahmad
2	Ali Khaled
3	Muna Ryan
4	Rena Waleed
5	Thamer Abdullah

Result Grid | Form Editor | Read Only

_Customers 1

Output

Action Output

#	Time	Action	Message
1	10:55:05	CREATE DATABASE PartyManagement	Error Code: 1007. Can't create database 'partymanagement'; datab
2	10:55:09	USE PartyManagement	0 row(s) affected
3	10:55:27	CREATE VIEW Simple_Customers AS SELECT * FROM Customerr	Error Code: 1050. Table 'Simple_Customers' already exists
4	10:55:31	SELECT * FROM Simple_Customers LIMIT 0, 1000	5 row(s) returned

Displays all columns from the Customers table